Trigonometric Equation: Questions and Shortened Solutions

Question 01

English

Let λ be a real number with $\lambda < -2$. Then the sum of all the principal solutions (that is, the solutions lying in $[0, 2\pi)$) of the equation $\tan x + \cot x = \lambda$ is equal to

(A) 3π
(B) 4π
(C) 5π
(D) 6π

Only one option is correct.

हिन्दी

माना कि λ एक वास्तविक संख्या है जिसके लिए $\lambda < -2$ है। तब समीकरण $\tan x + \cot x = \lambda$ के सभी मुख्य हलों (अर्थात् $[0, 2\pi)$ में स्थित हलों) का योग बराबर है

(A) 3π
(B) 4π
(C) 5π
(D) 6π

केवल एक विकल्प सही है।

Solution

$\tan x + \cot x = \dfrac{1}{\sin x \cos x} = \dfrac{2}{\sin 2x} = \lambda \Rightarrow \sin 2x = \dfrac{2}{\lambda} = -k$, where $0 < k < 1$

Let $\theta = \sin^{-1} k \in \left(0, \frac{\pi}{2}\right)$. General solution: $2x = n\pi + (-1)^n(-\theta)$

Principal solutions need $2x \in [0, 4\pi)$: $2x = \pi + \theta,\ 2\pi - \theta,\ 3\pi + \theta,\ 4\pi - \theta$ (none is a multiple of $\frac{\pi}{2}$, so $\tan x, \cot x$ exist)

Sum of the four values of $2x$ is 10π, so the sum of the principal solutions is 5π, whatever $\lambda < -2$ is

Correct Option: (C)

Question 02

English

The number of solutions of the equation $\tan^2 x + \cot^2 x - 3(\tan x - \cot x) = 2$ in the interval $(0, \pi)$ is

(A) 4
(B) 2
(C) 3
(D) 6

Only one option is correct.

हिन्दी

अन्तराल $(0, \pi)$ में समीकरण $\tan^2 x + \cot^2 x - 3(\tan x - \cot x) = 2$ के हलों की संख्या है

(A) 4
(B) 2
(C) 3
(D) 6

केवल एक विकल्प सही है।

Solution

Put $u = \tan x - \cot x$; then $\tan^2 x + \cot^2 x = u^2 + 2$ (as $\tan x \cot x = 1$)

Equation: $u^2 + 2 - 3u = 2 \Rightarrow u(u-3) = 0$

$u = 0$: $\tan^2 x = 1 \Rightarrow x = \frac{\pi}{4}, \ \frac{3\pi}{4}$

$u = 3$: $\tan^2 x - 3\tan x - 1 = 0$; product of roots -1, so one positive and one negative value of $\tan x$, each giving one x in $(0, \pi) \setminus \{\frac{\pi}{2}\}$

Total $2 + 2 = 4$ solutions

Correct Option: (A)

Question 03

English

Let S be the set of all $x \in [0, 2\pi]$, $x \neq \frac{\pi}{2}, \frac{3\pi}{2}$, such that $\tan x > 2\sin x$. Then S is a union of intervals whose total length is

(A) $\frac{\pi}{2}$
(B) π
(C) $\frac{4\pi}{3}$
(D) $\frac{2\pi}{3}$

Only one option is correct.

हिन्दी

माना कि S उन सभी $x \in [0, 2\pi]$, $x \neq \frac{\pi}{2}, \frac{3\pi}{2}$, का समुच्चय है जिनके लिए $\tan x > 2\sin x$ है। तब S अन्तरालों का एक सम्मिलन है जिनकी कुल लम्बाई है

(A) $\frac{\pi}{2}$
(B) π
(C) $\frac{4\pi}{3}$
(D) $\frac{2\pi}{3}$

केवल एक विकल्प सही है।

Solution

$\tan x - 2\sin x = \dfrac{\sin x\,(1 - 2\cos x)}{\cos x} > 0$

Sign changes at $0,\ \frac{\pi}{3},\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2},\ \frac{5\pi}{3},\ 2\pi$

$(0, \frac{\pi}{3})$: $(+)(-)/(+) < 0$; $(\frac{\pi}{3}, \frac{\pi}{2})$: > 0; $(\frac{\pi}{2}, \pi)$: < 0; $(\pi, \frac{3\pi}{2})$: $(-)(+)/(-) > 0$; $(\frac{3\pi}{2}, \frac{5\pi}{3})$: < 0; $(\frac{5\pi}{3}, 2\pi)$: $(-)(-)/(+) > 0$

$S = \left(\frac{\pi}{3}, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right)$, total length $\frac{\pi}{6} + \frac{\pi}{2} + \frac{\pi}{3} = \pi$

Correct Option: (B)

Question 04

English

Let S be the set of all $x \in [0, 2\pi]$ for which $\tan x$ and $\cot x$ are defined and which satisfy both the inequalities

$$2\sin^2 x - 3\sin x + 1 < 0 \quad \text{and} \quad \tan x > \cot x.$$

Then S is a union of intervals whose total length is

(A) $\frac{\pi}{4}$
(B) $\frac{5\pi}{12}$
(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{3}$

Only one option is correct.

हिन्दी

माना कि S उन सभी $x \in [0, 2\pi]$ का समुच्चय है जिनके लिए $\tan x$ तथा $\cot x$ परिभाषित हैं और जो दोनों असमिकाओं

$$2\sin^2 x - 3\sin x + 1 < 0$$

तथा

$$\tan x > \cot x$$

को सन्तुष्ट करते हैं। तब S अन्तरालों का एक सम्मिलन है जिनकी कुल लम्बाई है

(A) $\frac{\pi}{4}$

(B) $\frac{5\pi}{12}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{3}$

केवल एक विकल्प सही है।

Solution

$(2\sin x - 1)(\sin x - 1) < 0 \Rightarrow \frac{1}{2} < \sin x < 1 \Rightarrow x \in \left(\frac{\pi}{6}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$

$\tan x > \cot x \iff \frac{\tan^2 x - 1}{\tan x} > 0 \iff \tan x > 1$ or $-1 < \tan x < 0$

On $\left(\frac{\pi}{6}, \frac{\pi}{2}\right)$ $(\tan x > 0)$: $\tan x > 1 \Rightarrow x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

On $\left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$ $(\tan x < 0)$: $-1 < \tan x < 0 \Rightarrow x \in \left(\frac{3\pi}{4}, \frac{5\pi}{6}\right)$

Total length $\frac{\pi}{4} + \frac{\pi}{12} = \frac{\pi}{3}$

Correct Option: (D)

Question 05

English

The number of solutions of the equation $\tan(\pi \sin x) = 1$ in the interval $[0, 9\pi]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[0, 9\pi]$ में समीकरण $\tan(\pi \sin x) = 1$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

General solution of $\tan\theta = 1$: $\theta = n\pi + \frac{\pi}{4}$, so $\pi \sin x = n\pi + \frac{\pi}{4} \Rightarrow \sin x = n + \frac{1}{4}$

$|\sin x| \leq 1 \Rightarrow n = 0$ or $n = -1$: $\sin x = \frac{1}{4}$ or $\sin x = -\frac{3}{4}$ (not just the principal value $\frac{\pi}{4}$)

For these values $\pi \sin x \neq \pm\frac{\pi}{2}$, so $\tan(\pi \sin x)$ is defined

Each period $[2k\pi, 2k\pi + 2\pi)$: $\sin x = \frac{1}{4}$ twice (in $(2k\pi, 2k\pi + \pi)$), $\sin x = -\frac{3}{4}$ twice, so 4 solutions

$[0, 8\pi]$: $4 \times 4 = 16$; $[8\pi, 9\pi]$ has $\sin x \geq 0$: only $\sin x = \frac{1}{4}$, 2 solutions

Total $16 + 2 = 18$

Final Answer: 18

Question 06

English

Let S be the set of all $x \in (0, 2\pi)$ satisfying the equation

$$3\sin 2x - 4\cos 2x = 8\sin x - 4\cos x.$$

Then the value of $\sin\left(\sum_{x\in S} x\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी $x \in (0, 2\pi)$ का समुच्चय है जो समीकरण

$$3\sin 2x - 4\cos 2x = 8\sin x - 4\cos x$$

को सन्तुष्ट करते हैं। तब $\sin\left(\sum_{x\in S} x\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$3\sin 2x - 4\cos 2x - 8\sin x + 4\cos x = 4\sin^2 x + 6\sin x\cos x - 4\cos^2 x - 8\sin x + 4\cos x$
$= 2(2\sin x - \cos x)(\sin x + 2\cos x) - 4(2\sin x - \cos x) = 2(2\sin x - \cos x)(\sin x + 2\cos x - 2)$
$2\sin x = \cos x \Rightarrow \tan x = \frac{1}{2}$: $x = \alpha,\ \pi + \alpha$, where $\tan\alpha = \frac{1}{2}$, $0 < \alpha < \frac{\pi}{2}$
$\sin x + 2\cos x = 2$: with $t = \tan\frac{x}{2}$, $2t + 2 - 2t^2 = 2 + 2t^2 \Rightarrow t = 0$ or $t = \frac{1}{2}$; $t = 0$ gives $x = 0 \notin S$, $t = \frac{1}{2}$ gives $x = \beta$ with $\tan\beta = \frac{4}{3}$, $\beta = 2\alpha$
Sum $= \alpha + (\pi + \alpha) + 2\alpha = \pi + 4\alpha$, so $\sin(\pi + 4\alpha) = -\sin 4\alpha = -2\sin 2\alpha\cos 2\alpha = -2\cdot\frac{4}{5}\cdot\frac{3}{5} = -0.96$

Final Answer: -0.96

Question 07

English

The sum of all solutions of the equation

$$\sin 3x - \sin x + \sqrt{3}\,(\cos 3x + \cos x) = 2\cos 2x$$

lying in the interval $[-\pi, 0]$ is $\dfrac{k\pi}{6}$. Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[-\pi, 0]$ में स्थित समीकरण

$$\sin 3x - \sin x + \sqrt{3}\,(\cos 3x + \cos x) = 2\cos 2x$$

के सभी हलों का योग $\dfrac{k\pi}{6}$ है। तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\sin 3x - \sin x = 2\cos 2x \sin x$, $\cos 3x + \cos x = 2\cos 2x \cos x$

Equation: $2\cos 2x\left(\sin x + \sqrt{3}\cos x - 1\right) = 0$

$\cos 2x = 0 \Rightarrow x = \frac{\pi}{4} + \frac{n\pi}{2}$; in $[-\pi, 0]$: $-\frac{\pi}{4},\ -\frac{3\pi}{4}$

$\sin x + \sqrt{3}\cos x = 1 \Rightarrow \sin\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$; $x + \frac{\pi}{3} \in \left[-\frac{2\pi}{3}, \frac{\pi}{3}\right]$ gives only $x + \frac{\pi}{3} = \frac{\pi}{6}$, i.e. $x = -\frac{\pi}{6}$

Sum $= -\frac{\pi}{4} - \frac{3\pi}{4} - \frac{\pi}{6} = -\frac{7\pi}{6} \Rightarrow k = -7$

Final Answer: -7

Question 08

English

Let S be the set of all $x \in (0, 2\pi)$, $x \neq \pi$, satisfying the equation $4\sin x - 3\cos x - 3 = \tan\dfrac{x}{2}$. Then the value of $\displaystyle\sum_{x\in S} \sin x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी $x \in (0, 2\pi)$, $x \neq \pi$, का समुच्चय है जो समीकरण $4\sin x - 3\cos x - 3 = \tan\dfrac{x}{2}$ को सन्तुष्ट करते हैं। तब $\displaystyle\sum_{x\in S} \sin x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $t = \tan\frac{x}{2}$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$; for $x \in (0, 2\pi) \setminus \{\pi\}$, t takes each real value once

$8t - 3(1 - t^2) - 3(1 + t^2) = t(1 + t^2) \Rightarrow 8t - 6 = t + t^3 \Rightarrow t^3 - 7t + 6 = 0$

$(t - 1)(t - 2)(t + 3) = 0 \Rightarrow t = 1,\ 2,\ -3$: three solutions

$\sin x = \frac{2t}{1+t^2}$: $1,\ \frac{4}{5},\ -\frac{3}{5}$; sum $= 1 + 0.8 - 0.6 = 1.2$

Final Answer: 1.2

Question 09

English

Let x be a real number satisfying $\log_{\cos x}(6\cos x - 1) = \dfrac{1}{2}$. Then the value of $\cos x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x एक वास्तविक संख्या है जो $\log_{\cos x}(6\cos x - 1) = \dfrac{1}{2}$ को सन्तुष्ट करती है। तब $\cos x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Domain: base $0 < \cos x$, $\cos x \neq 1$; argument $6\cos x - 1 > 0$

Equation $\iff 6\cos x - 1 = \sqrt{\cos x}$; with $c = \cos x$, squaring: $36c^2 - 12c + 1 = c \Rightarrow 36c^2 - 13c + 1 = 0$

$c = \frac{13 \pm 5}{72} = \frac{1}{4}$ or $\frac{1}{9}$

$c = \frac{1}{9}$: $6c - 1 = -\frac{1}{3} < 0$, extraneous (and outside the domain); $c = \frac{1}{4}$: $6c - 1 = \frac{1}{2} = \sqrt{\frac{1}{4}}$ and $0 < c < 1$, valid

$\cos x = 0.25$

Final Answer: 0.25

Question 10

English

The number of solutions of the equation $\sqrt{2 - 2\sin 3x}\,\sec 3x = 2$ in the interval $[0, 100\pi]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[0, 100\pi]$ में समीकरण $\sqrt{2 - 2\sin 3x}\,\sec 3x = 2$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\theta = 3x \in [0, 300\pi]$. Need $\cos\theta \neq 0$; the equation is $\sqrt{2 - 2\sin\theta} = 2\cos\theta$, so $\cos\theta > 0$

Squaring: $2 - 2s = 4(1 - s^2)$ with $s = \sin\theta \Rightarrow 2s^2 - s - 1 = 0 \Rightarrow s = 1$ or $s = -\frac{1}{2}$

$s = 1$: $\cos\theta = 0$, $\sec\theta$ undefined, rejected (it satisfies the equation without sec)

$s = -\frac{1}{2}$ with $\cos\theta > 0$: $\theta = \frac{11\pi}{6} + 2n\pi$ ($\theta = \frac{7\pi}{6} + 2n\pi$ is extraneous, $\cos\theta < 0$)

$0 \leq \frac{11\pi}{6} + 2n\pi \leq 300\pi \Rightarrow 0 \leq n \leq 149$: 150 values of θ, hence 150 values of x

Final Answer: 150

Question 11

English

Consider $f(x) = \sqrt{-\tan x} + \log_e\left(4\pi^2 - x^2\right)$. Let D be the set of all real numbers x for which $f(x)$ is defined, and let $S = \{x \in D : \sin 2x + \sin x = 0\}$. If the sum of all elements of S is $\frac{k\pi}{3}$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$f(x) = \sqrt{-\tan x} + \log_e\left(4\pi^2 - x^2\right)$ पर विचार कीजिए। माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $f(x)$ परिभाषित है, तथा $S = \{x \in D : \sin 2x + \sin x = 0\}$ है। यदि S के सभी अवयवों का योग $\frac{k\pi}{3}$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

D: $4\pi^2 - x^2 > 0 \Rightarrow -2\pi < x < 2\pi$; $\tan x$ defined and $\tan x \le 0$

$\sin 2x + \sin x = \sin x(2\cos x + 1) = 0 \Rightarrow \sin x = 0$ or $\cos x = -\frac{1}{2}$

$\sin x = 0$ in $(-2\pi, 2\pi)$: $-\pi, 0, \pi$, all with $\tan x = 0 \le 0$, so all in S ($\pm 2\pi$ excluded by the log)

$\cos x = -\frac{1}{2}$: $\pm\frac{2\pi}{3}, \pm\frac{4\pi}{3}$; $\tan\frac{2\pi}{3} = -\sqrt{3}$ and $\tan\left(-\frac{4\pi}{3}\right) = -\sqrt{3}$ kept; $\tan\left(-\frac{2\pi}{3}\right) = \tan\frac{4\pi}{3} = \sqrt{3}$ rejected

Sum $= -\pi + 0 + \pi + \frac{2\pi}{3} - \frac{4\pi}{3} = -\frac{2\pi}{3} \Rightarrow k = -2$

Final Answer: -2

Question 12

English

The number of solutions of the equation $\cos(\pi x) = \frac{x^2}{36}$ in the interval $[-5, 6]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[-5, 6]$ में समीकरण $\cos(\pi x) = \frac{x^2}{36}$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Graphs: $y = \cos\pi x$ (period 2, value 1 at even integers, -1 at odd integers) and $y = \frac{x^2}{36} \in [0, 1]$ on $[-6, 6]$

On each unit interval $[m, m+1]$ the cosine runs monotonically between -1 and 1 while $\frac{x^2}{36} < 1$ for $|x| < 6$, so exactly one crossing

$[-5, 0]$: one on each of $[-5, -4], \ldots, [-1, 0]$: 5 ($x = 0$ is not a root)

$[0, 6]$: one in each of $(0, 1), \ldots, (4, 5)$: 5; on $[5, 6]$: a crossing in $(5, 6)$ and also $x = 6$ ($\cos 6\pi = 1 = \frac{36}{36}$): 2

Total $5 + 5 + 2 = 12$

Final Answer: 12

Question 13

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \max\{\sin|x|,\ \cos|x|\}$. Let S be the set of all $x \in [-2\pi, \pi]$ such that $f(x) = -\dfrac{1}{2}$. If the sum of all elements of S is $k\pi$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \max\{\sin|x|,\ \cos|x|\}$ द्वारा परिभाषित है। माना कि S उन सभी $x \in [-2\pi, \pi]$ का समुच्चय है जिनके लिए $f(x) = -\dfrac{1}{2}$ है। यदि S के सभी अवयवों का योग $k\pi$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f(x) = g(|x|)$ with $g(t) = \max\{\sin t, \cos t\}$; for $t \in [0, 2\pi]$: $g = \sin t$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$, $g = \cos t$ otherwise
$t \in [0, \pi]$: $\sin t \geq 0$, so $g(t) \geq 0$; hence no $x \in [0, \pi]$ is in S
$x \in [-2\pi, 0)$: $t = -x \in (0, 2\pi]$. Line $y = -\frac{1}{2}$ meets g at $t = \frac{7\pi}{6}$ (sine branch, as $\frac{7\pi}{6} < \frac{5\pi}{4}$) and $t = \frac{4\pi}{3}$ (cosine branch, as $\frac{4\pi}{3} > \frac{5\pi}{4}$)
$t = \frac{11\pi}{6}$ ($\sin = -\frac{1}{2}$) and $t = \frac{2\pi}{3}$ ($\cos = -\frac{1}{2}$) are not points of $g = -\frac{1}{2}$, since there the other function is larger
$S = \left\{-\frac{7\pi}{6}, -\frac{4\pi}{3}\right\}$, sum $-\frac{5\pi}{2} \Rightarrow k = -2.5$

Final Answer: -2.5

Question 14

English

The sum of all real numbers x satisfying the equation

$$4^{\sin^2 \pi x} + 4^{\cos^2 \pi x} = 4 - |4x - 3|$$

is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समीकरण

$$4^{\sin^2 \pi x} + 4^{\cos^2 \pi x} = 4 - |4x - 3|$$

को सन्तुष्ट करने वाली सभी वास्तविक संख्याओं x का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

AM $\geq$ GM: $4^{\sin^2 \pi x} + 4^{\cos^2 \pi x} \geq 2\sqrt{4^{\sin^2 \pi x + \cos^2 \pi x}} = 4$, equality iff $\sin^2 \pi x = \cos^2 \pi x = \frac{1}{2}$
Right side $4 - |4x - 3| \leq 4$, equality iff $x = \frac{3}{4}$
So both sides must equal 4: $x = \frac{3}{4}$, and then $\sin^2 \frac{3\pi}{4} = \frac{1}{2}$ holds
Unique solution $x = \frac{3}{4}$; sum $= 0.75$

Final Answer: 0.75

Question 15

English

The number of solutions of the equation $\sqrt{3}\sin x + \cos x + \sin\dfrac{3x}{2} = 3$ in the interval $[0, 500\pi]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[0, 500\pi]$ में समीकरण $\sqrt{3}\sin x + \cos x + \sin\dfrac{3x}{2} = 3$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\sqrt{3}\sin x + \cos x = 2\sin\left(x + \frac{\pi}{6}\right) \le 2$ and $\sin\frac{3x}{2} \le 1$, so the left side ≤ 3
Equality needs both: $\sin\left(x + \frac{\pi}{6}\right) = 1 \Rightarrow x = \frac{\pi}{3} + 2n\pi$, and $\sin\frac{3x}{2} = 1$
At $x = \frac{\pi}{3} + 2n\pi$: $\frac{3x}{2} = \frac{\pi}{2} + 3n\pi$, $\sin = \cos 3n\pi = (-1)^n$, so n must be even
Solutions $x = \frac{\pi}{3} + 4m\pi$; $0 \le \frac{1}{3} + 4m \le 500 \Rightarrow 0 \le m \le 124$: 125 solutions

Final Answer: 125

Question 16

English

Let x be a real number, not an integral multiple of $\dfrac{\pi}{2}$, such that

$$\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} + \frac{4}{5} = 0.$$

Then the value of $\tan^2 x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x एक वास्तविक संख्या है, जो $\dfrac{\pi}{2}$ का पूर्णांक गुणज नहीं है, जिसके लिए

$$\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} + \frac{4}{5} = 0$$

है। तब $\tan^2 x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\frac{\sin 3x}{\sin x} = 3 - 4\sin^2 x$, $\frac{\cos 3x}{\cos x} = 4\cos^2 x - 3$; sum $= 4(\cos^2 x - \sin^2 x) = 4\cos 2x$
$4\cos 2x = -\frac{4}{5} \Rightarrow \cos 2x = -\frac{1}{5}$ (possible, and then $\sin x, \cos x \neq 0$)

$$\tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x} = \frac{6/5}{4/5} = 1.5$$

Final Answer: 1.5

Question 17

English

Let S be the set of all $x \in (-2\pi, 2\pi)$ for which $\sec x$, $\tan x$ and $\tan\dfrac{x}{2}$ are defined and

$$\sec x + \tan x = 1 + 4\tan\frac{x}{2}.$$

Then the value of $\displaystyle\sum_{x\in S}(\sin x + \cos x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी $x \in (-2\pi, 2\pi)$ का समुच्चय है जिनके लिए $\sec x$, $\tan x$ तथा $\tan\dfrac{x}{2}$ परिभाषित हैं और

$$\sec x + \tan x = 1 + 4\tan\frac{x}{2}$$

है। तब $\displaystyle\sum_{x\in S}(\sin x + \cos x)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $t = \tan\frac{x}{2}$: $\sec x + \tan x = \dfrac{1+\sin x}{\cos x} = \dfrac{(1+t)^2}{1-t^2} = \dfrac{1+t}{1-t}$ $(t \neq \pm 1)$

$\dfrac{1+t}{1-t} = 1 + 4t \Rightarrow 1 + t = 1 + 3t - 4t^2 \Rightarrow 2t(2t-1) = 0 \Rightarrow t = 0$ or $t = \frac{1}{2}$

$t = 0$: $x = 0$ ($\pm 2\pi$ excluded); $\sin x + \cos x = 1$

$t = \frac{1}{2}$: $\sin x = \frac{4}{5}$, $\cos x = \frac{3}{5}$; $\frac{x}{2} \in (-\pi, \pi)$ gives $\frac{x}{2} = \tan^{-1}\frac{1}{2}$ or $\tan^{-1}\frac{1}{2} - \pi$: two values of x, each with $\sin x + \cos x = \frac{7}{5}$

Sum $= 1 + \frac{7}{5} + \frac{7}{5} = 3.8$

Final Answer: 3.8

Question 18

English

Let x and y be real numbers such that

$$x - y = \frac{\pi}{2} \qquad \text{and} \qquad 3\sin x - 4\sin y = 5.$$

Then the value of $\sin(x + y)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x तथा y वास्तविक संख्याएँ हैं जिनके लिए

$$x - y = \frac{\pi}{2}$$

तथा

$$3\sin x - 4\sin y = 5$$

हैं। तब $\sin(x+y)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y = x - \frac{\pi}{2} \Rightarrow \sin y = -\cos x$, so $3\sin x + 4\cos x = 5$

$3\sin x + 4\cos x \leq \sqrt{3^2+4^2} = 5$, equality only when $\sin x = \frac{3}{5}$, $\cos x = \frac{4}{5}$

$x + y = 2x - \frac{\pi}{2} \Rightarrow \sin(x+y) = -\cos 2x = -(\cos^2 x - \sin^2 x) = -\frac{16-9}{25} = -0.28$

Final Answer: -0.28